

Mark Park

Lead Software Engineer

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SUMMARY

Lead Software Engineer with 4+ years on Franklin Templeton Wealth Management Technology's client onboarding and money-movement platforms. Backend lead of the Account Lifecycle Platform (AOP 2.0) rebuild for the High Net Worth business — a 5-engineer team delivering account opening, KYC/AML screening (LexisNexis), DocuSign e-signature workflows, Drools rules, and a dynamic workflow orchestration engine I designed to replace Camunda. Earlier a core builder and one of two primary production maintainers of CashApp, the firm's money-movement platform. Java/Spring microservices and Angular on Azure (Cosmos DB, SQL Server, Snowflake); established an AI-First, TDD-gated agent-based code generation practice on GitHub Copilot CLI adopted across three teams.

TECHNICAL SKILLS

Languages: Java 8/17/21 (led multi-version migrations), Python, SQL, TypeScript/JavaScript, C++

Backend, Architecture & Full-Stack: Spring Boot, Spring Cloud, Spring Security, Spring Batch, Spring WebFlux, Hibernate/JPA, REST & gRPC API design, Microservices, Domain-Driven Design, Event-Driven Architecture, SAGA Choreography, Angular, React

Domain Expertise — Client Onboarding & Payments: Workflow orchestration engines (Camunda/BPMN, custom-built), Drools rules engine (BRMS), KYC/AML and sanctions screening (LexisNexis), DocuSign e-signature workflows, ACH/wire/check processing (NACHA), OFAC screening and AML compliance, regulatory audit trails

Cloud, Data & DevOps: Azure (Cosmos DB, Service Bus, Event Grid, Key Vault, Entra ID), AWS (EC2, S3, ECS), Kubernetes, Docker, Kafka, RabbitMQ, Redis, SQL Server, Snowflake, MongoDB; CI/CD (GitHub Actions, Azure Pipelines, Jenkins), SonarQube, Datadog

AI-First Engineering: TDD-gated, agent-reviewed code generation; GitHub Copilot CLI and GitHub MCP; prompt and loop engineering; Claude Code

Testing & Process: JUnit, Mockito, Postman, automated regression and API testing, Agile/Scrum, Jira, root cause analysis, incident management

PROFESSIONAL EXPERIENCE

Brillio — Client: Franklin Templeton (Fiduciary Trust) | Remote (Eastern hours) — New York, NY

Contract Software Engineer (May 2022 – Jun 2025) → Software Development Lead, Brillio full-time (Jun 2025 – Present)

Embedded 4+ years on Franklin Templeton Wealth Management Technology teams; performance earned a two-year contract extension, conversion to full-time Brillio employee, and promotion to Software Development Lead, with recognition from directors, architects, and other teams' leads for the AOP 2.0, SWP Forms, and CashApp releases. California-based on Eastern hours; open to a three-day hybrid schedule at a Franklin Templeton location.

Lead Software Engineer — Account Lifecycle Platform (AOP 2.0, High Net Worth Client Onboarding) | Aug 2024 – Present

- Founding engineer on the rebuild; promoted to lead backend engineering on the recommendation of senior architects and management. Led up to 5 engineers taking a heavily criticized legacy onboarding platform from POC to MVP to production — a release leadership recognized as a turnaround.
- Automated over 60% of previously manual onboarding and KYC logic and improved responsiveness 60–80% over the legacy platform, reducing manual touchpoints and improving operational efficiency for onboarding, compliance, and operations teams.
- Designed and built the platform's dynamic workflow orchestration engine (Spring Boot + SQL Server). Built the Camunda POC end-to-end with the IT manager; when a major Camunda upgrade broke the code months before MVP, rebuilt the entire workflow as a custom engine in one month.
- Its orchestration APIs cover every account-opening scenario — request, validation, assignment, Front Office / Retirement Services approval, KYC approval, DocuSign signature, SWP submission approval — with token- and role-based authorization against request status and sub-status, retries with backoff, and fault tolerance so no request ends in a broken state; new steps and approval paths are configured in the database without code changes.
- Architected the two core microservices along clear domain boundaries: a Drools-based rules engine governing all UI field behavior, with dynamic completion validation synced to the Angular UI for real-time, millisecond feedback, and a secure persistence service as the sole authorized gateway to internal Franklin Templeton systems.
- Delivered the document collection and e-signature workflow: UI and backend validation of required documents by account type and client, document storage APIs on Azure Blob, and production support of the DocuSign integration, including resolving live e-signature incidents.
- Built the KYC/AML screening API against LexisNexis from vendor documentation — a custom search returning screening hits for compliance review — as the POC for the onboarding KYC integration; contributing to the Instant ID evaluation and assisting the downstream IPS and KYC applications.

- Owned the data layer: Cosmos DB primary store (generic recursive Patch-API save utility, 5–10 ms, working around a known SDK upsert limitation), SQL Server workflow state, and Snowflake. Built a Python/Streamlit application on Snowflake that converts deprecated AOP 1.0 data into business-ready documents, used weekly by the business to review and migrate legacy accounts.
- Established an AI-First, TDD-gated agent-based code generation workflow on GitHub Copilot CLI: test suites written from business requirements first, AI agents implement against them, and every change is validated by agent-driven code review and agent-authored pull requests via GitHub MCP before it reaches production — existing functional tests are never broken. Self-taught the workflows on Claude Code on personal time, then translated them onto the firm-approved Copilot CLI.
- Led AI adoption across three product teams: authored the shared copilot-instructions.md, Copilot CLI commands integrated with Jira, and a Python database-alignment tool adopted by other projects; ran knowledge-transfer sessions on the CLI approach (prompt and loop engineering, reducing token usage, optimizing agent efficiency) for all 5 AOP 2.0 engineers plus the CashApp and SWP Forms leads.
- Top 5% firm-wide on Copilot cost-efficiency since the Jan 2026 rollout; 30K+ lines of AI-assisted code in production.
- Automated testing of onboarding flows with JUnit and Mockito; partnered with TQA on workflow-scenario automation scripts. Led the Java 17→21 migration ahead of MVP; sole backend on-call during market hours; grew into full-stack delivery of Angular features; mentored a summer intern to shipping production APIs.

Backend Engineer — SWP Forms (Systematic Withdrawal Plan) | Aug 2024 – Dec 2025 (concurrent with AOP 2.0)

- Hand-picked by the lead architect to build a standalone Drools rules-engine microservice from concept through MVP in 3 months; cut processing time for 500+ business rules from several seconds to 50–100 ms using fact injection, agenda groups, and salience-based prioritization. Later extended to a second production application; after handoff, onboarded the successor engineer and resolved production issues through Dec 2025.

Backend Developer — CashApp (Wealth Management money-movement platform: ACH, Wires, Check) | May 2022 – Aug 2024

- Core contributor from POC through MVP to production on the High Net Worth business's money-movement platform: ACH, wire, and check processing, NACHA, wire, and positive-pay file generation and parsing, and Axway and Azure Blob file integrations.
- One of two primary production maintainers overseeing live-market-hours processing of tens of millions of dollars daily (\$106B+ lifetime) against strict settlement cut-offs — never rolled to the next business day; owned root cause analysis of production incidents against SLAs and TQA release sign-off. Owned OFAC screening, AML monitoring, compliance-reconciliation, and audit-trail processing; built FIS Global Plus accounting, Cash Agent, external payment, and Azure check-reading AI integrations.
- Hand-built the platform's Spring/SQL Server transaction workflows and designed a scalable event-driven architecture on Azure Event Grid powering 20+ monitored scheduled batch jobs, several on 5-minute cycles. Eliminated production deadlocks (2–5 per day to zero) via optimistic concurrency control, retryable services, and custom thread pool executors; SAGA choreography over RabbitMQ, later Azure Service Bus.
- Delivered on Azure with Kubernetes-based deployments, GitLab/Bamboo CI/CD, regression and API test automation, and ServiceNow incident and change tracking; led design discussions across cross-functional groups of 20–30; later assisted the team's Java 8→21 upgrade.

EDUCATION & CERTIFICATIONS

San Diego State University — Bachelor of Science, Computer Science

Certifications in progress: Anthropic Claude Certified Developer – Foundations; Microsoft Azure AI & Agents Developer Associate